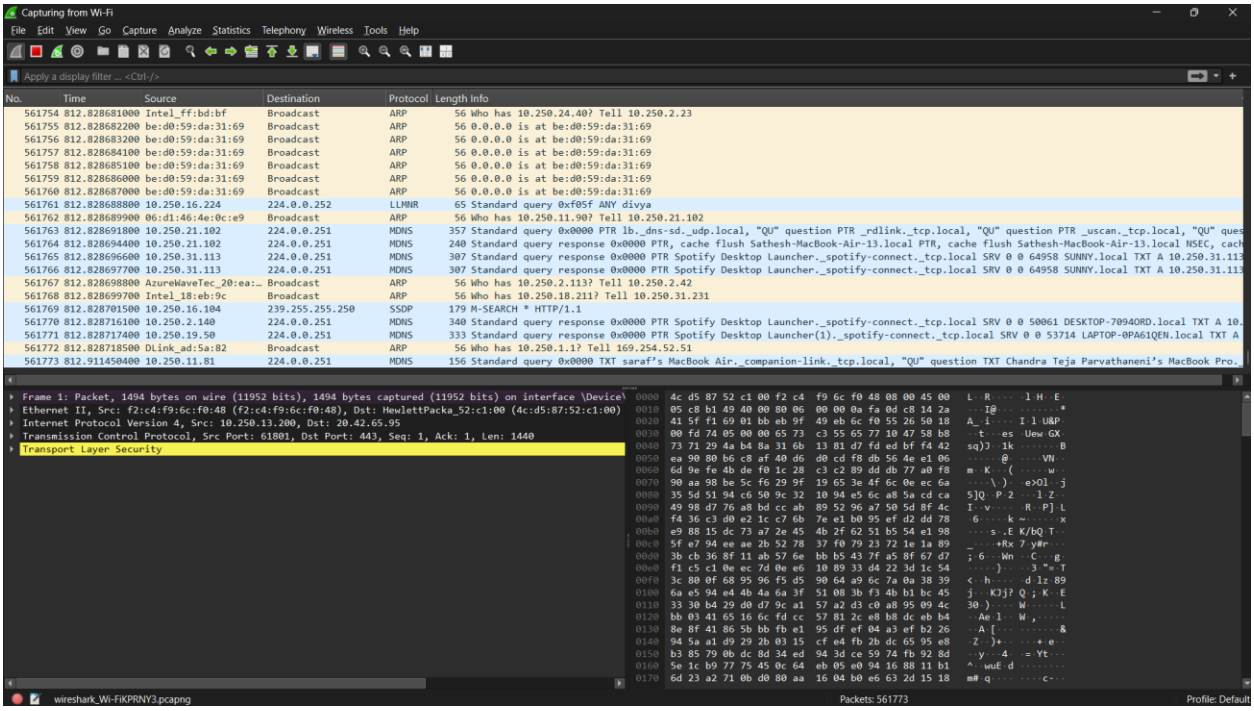
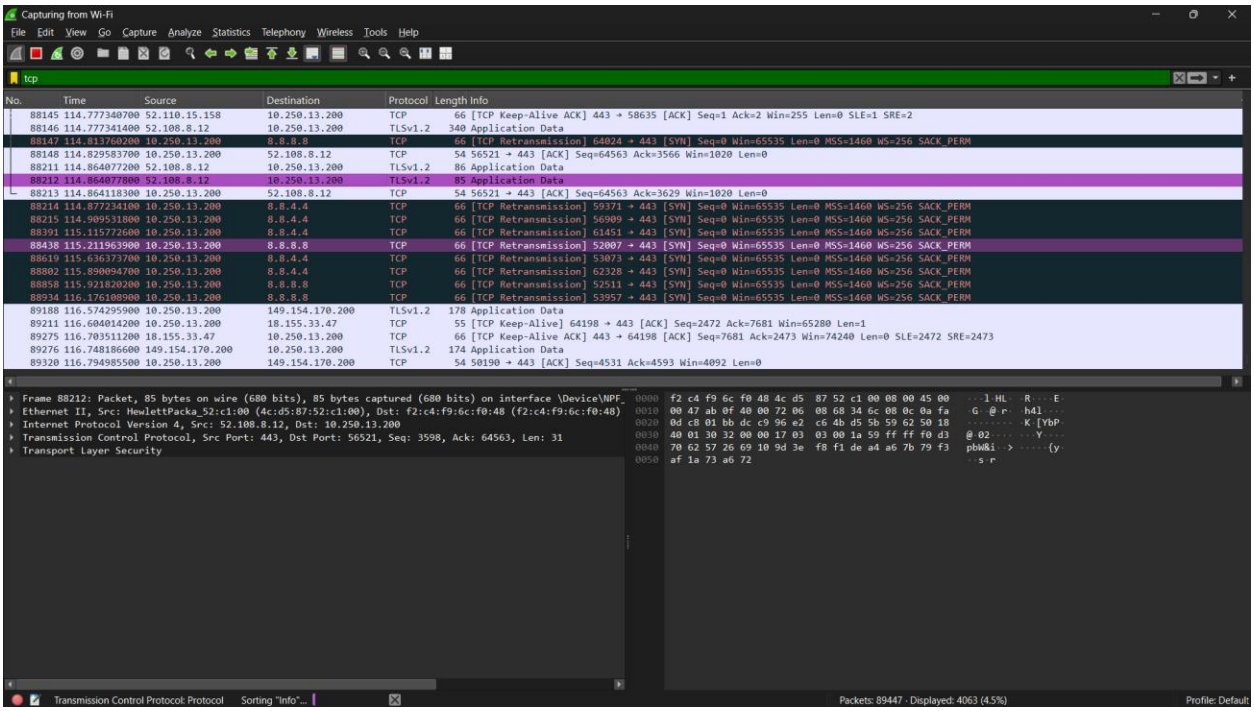
TASK 1: Packet Identification:-

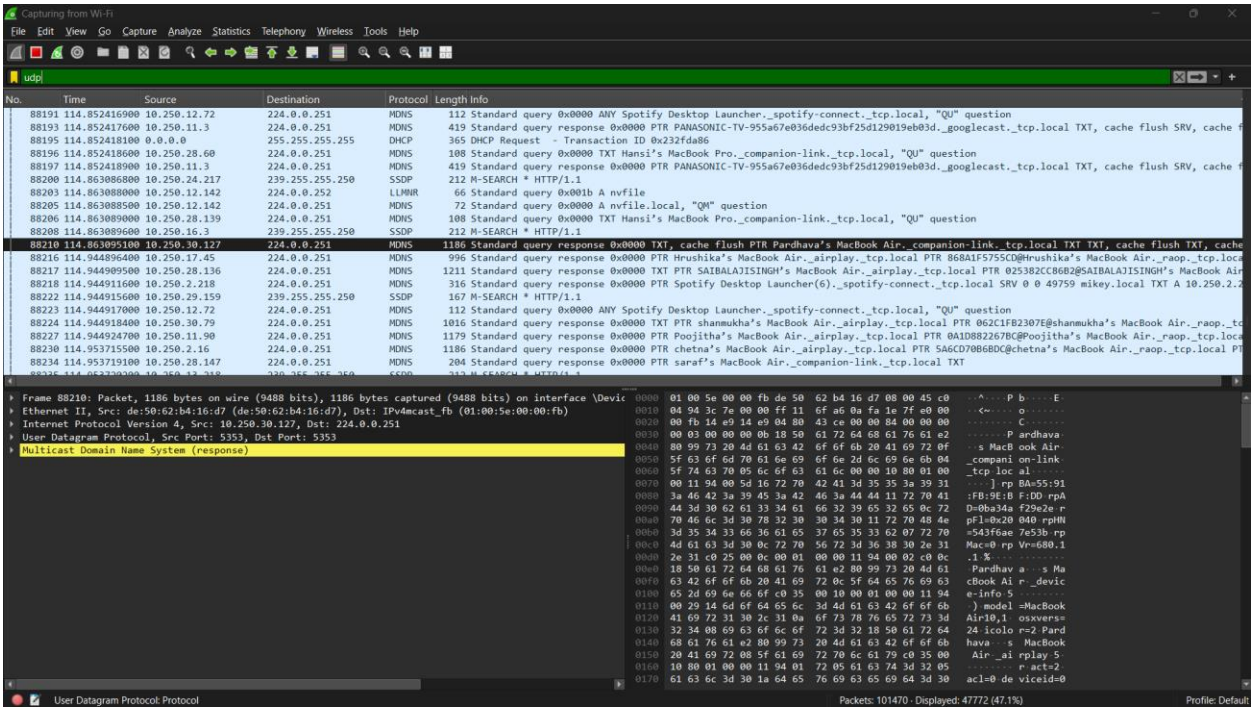
1. Open the PCAP file and identify the total number of packets captured.



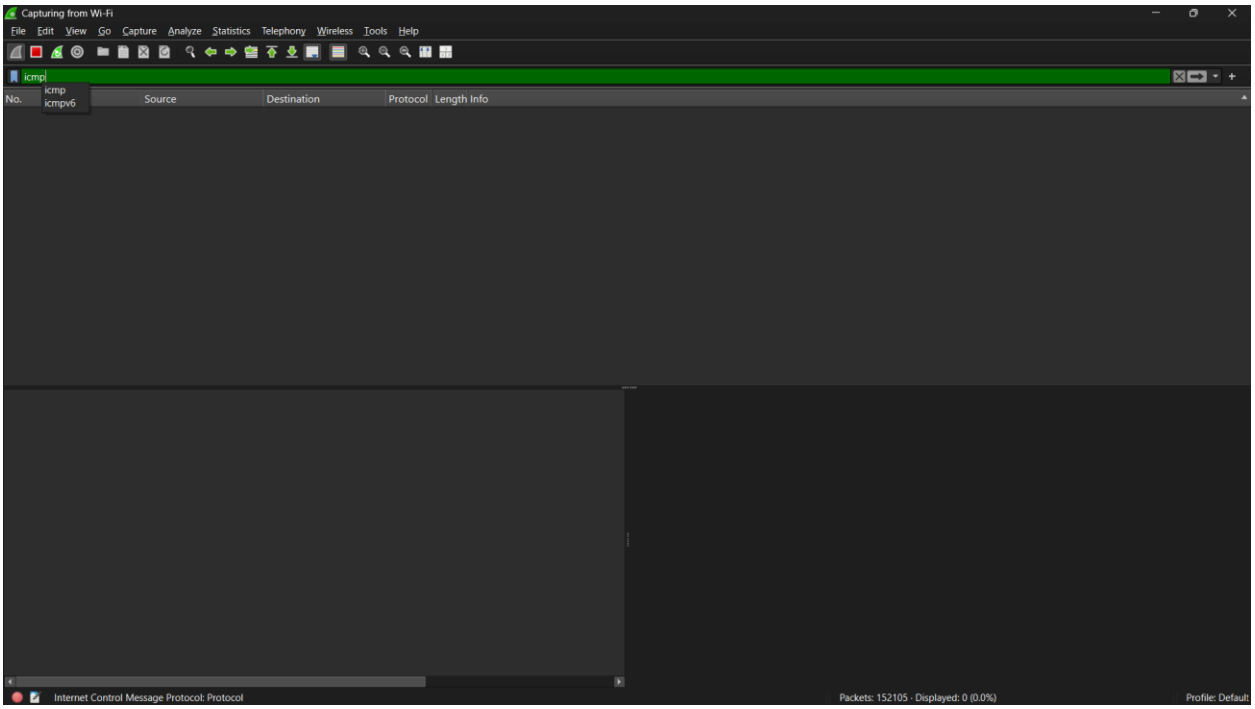
2.Apply the filter tcp and record the number of TCP packets



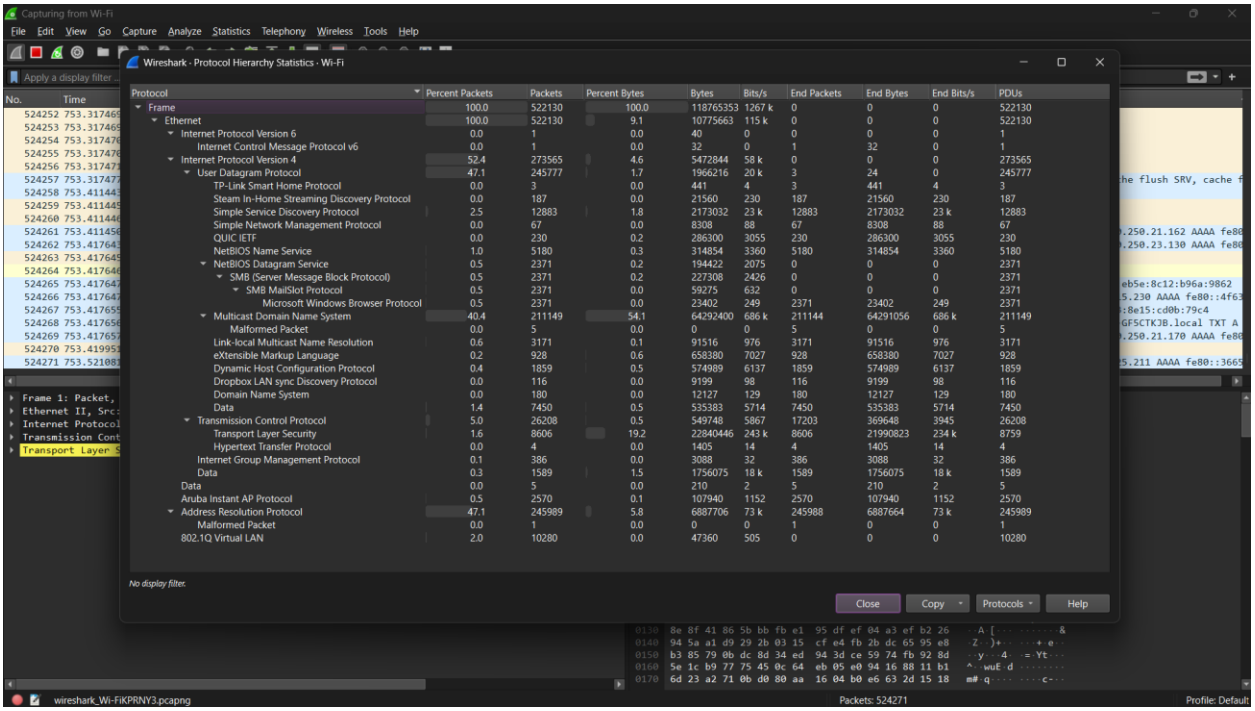
3.Apply the filter udp and record the number of UDP packets



4. Apply the filter icmp and record the number of ICMP packets.

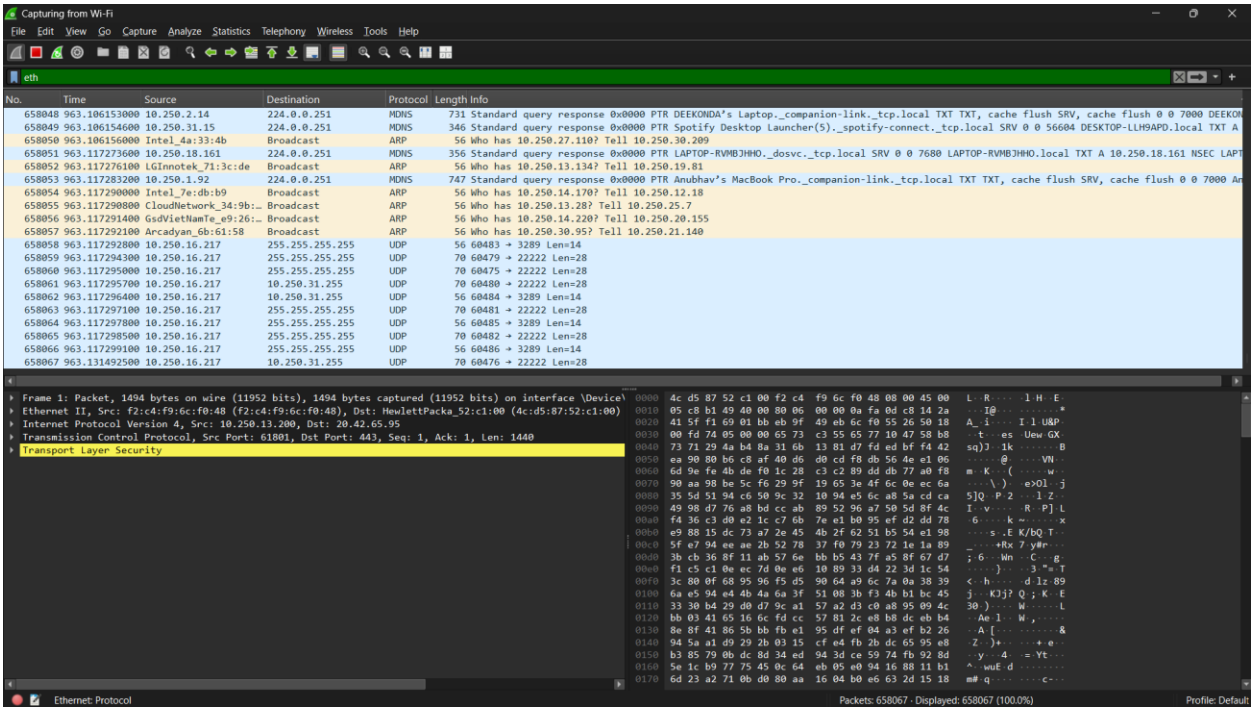


5. Identify the top five protocols using Statistics → Protocol Hierarchy.



Task 2: Ethernet frame inspection:

6. Apply the filter eth.

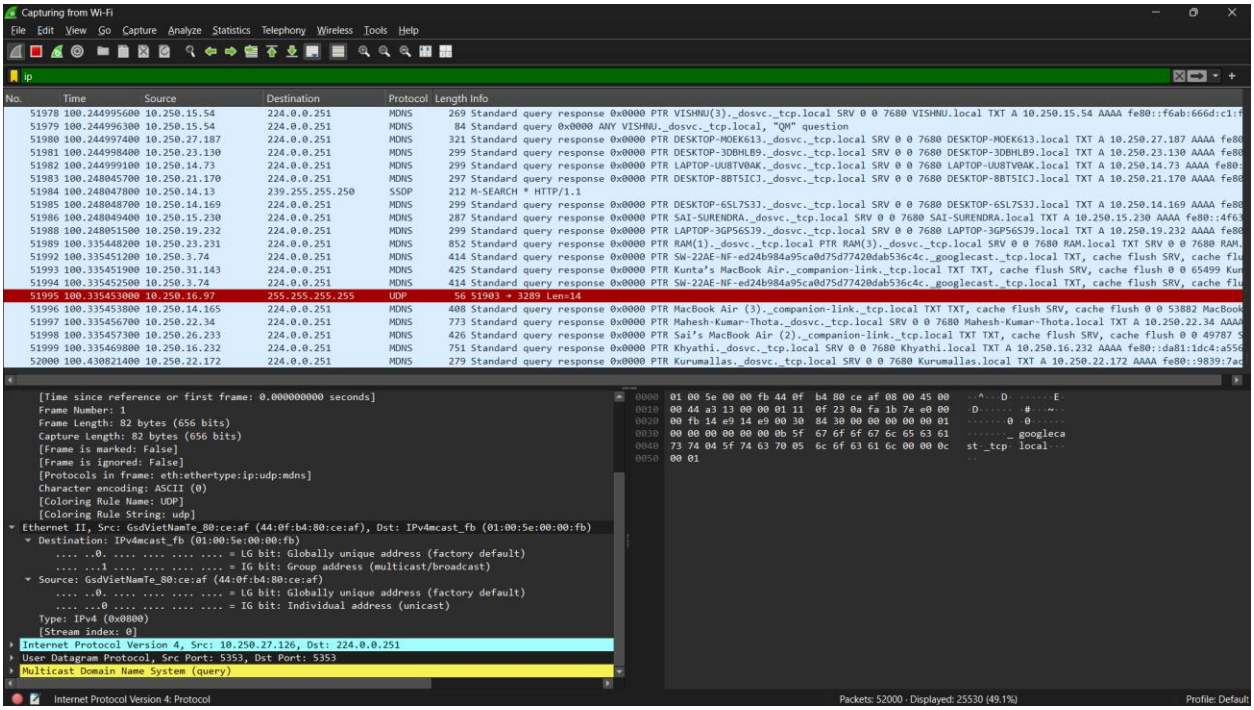


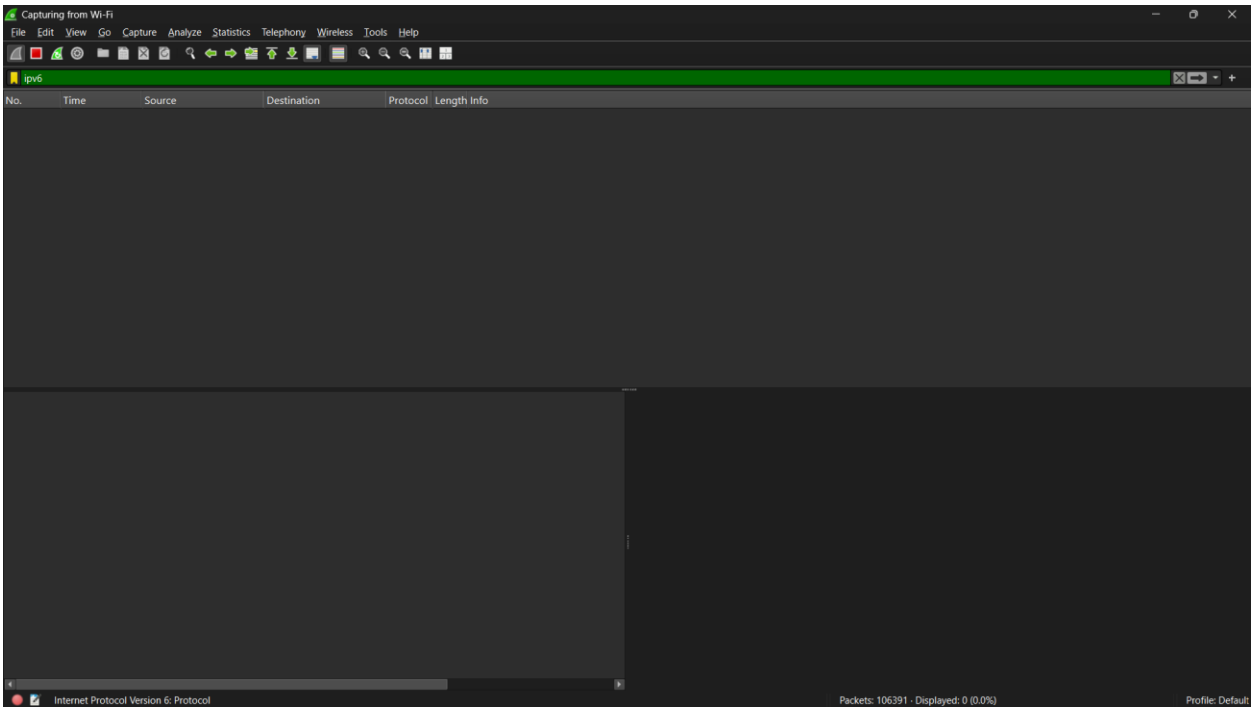
7. Select five different packets and record their: - Frame number - Frame length - Source MAC address - Destination MAC address – EtherType

Packet	Frame No.	Frame Length	Source MAC	Destination MAC	EtherType
1	751857	56 bytes	Source: CloudNetwork_e0:fc:3f (d8:b3:2f:e0:fc:3f)	Broadcast (ff:ff:ff:ff:ff:ff)	ARP
2	751864	531 bytes	Source: 26:27:01:d8:22:3f (26:27:01:d8:22:3f)	IPv4mcast_fb (01:00:5e:00:00:fb)	IPv4
3	751870	74 bytes	Source: AzureWaveTec_ff:33:6e (9c:c7:d3:ff:33:6e)	Destination: IPv4mcast_fc (01:00:5e:00:00:fc)	IPv4

4	751867	1117 bytes	Source: Intel_12:da:72 (40:c7:3c:12:da:72)	Destination: IPv4mcast_7f:ff:fa (01:00:5e:7f:ff:fa)	IPv4
5	751860	56 bytes	Source: CloudNetwork_52:d7:21 (3c:ef:a5:52:d7:21)	Destination: Broadcast (ff:ff:ff:ff:ff:ff)	ARP

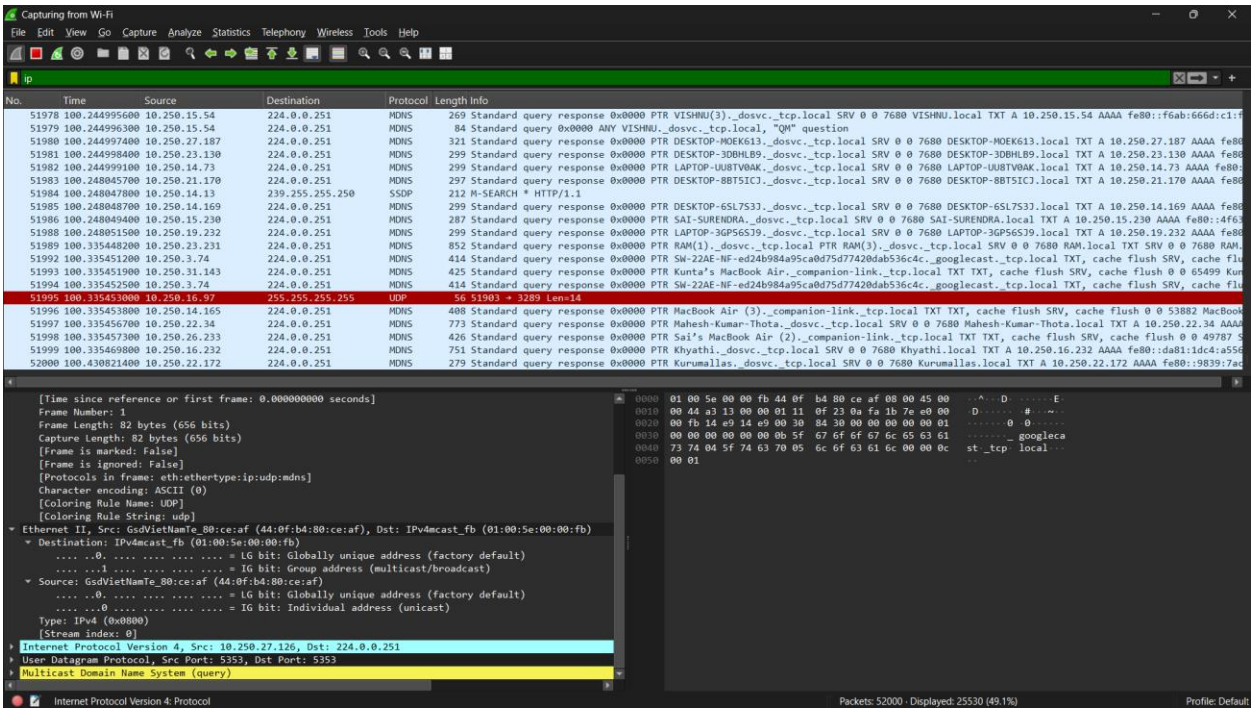
8. Identify packets carrying IPv4 and IPv6 traffic.





Task 3: IPv4 packet inspection

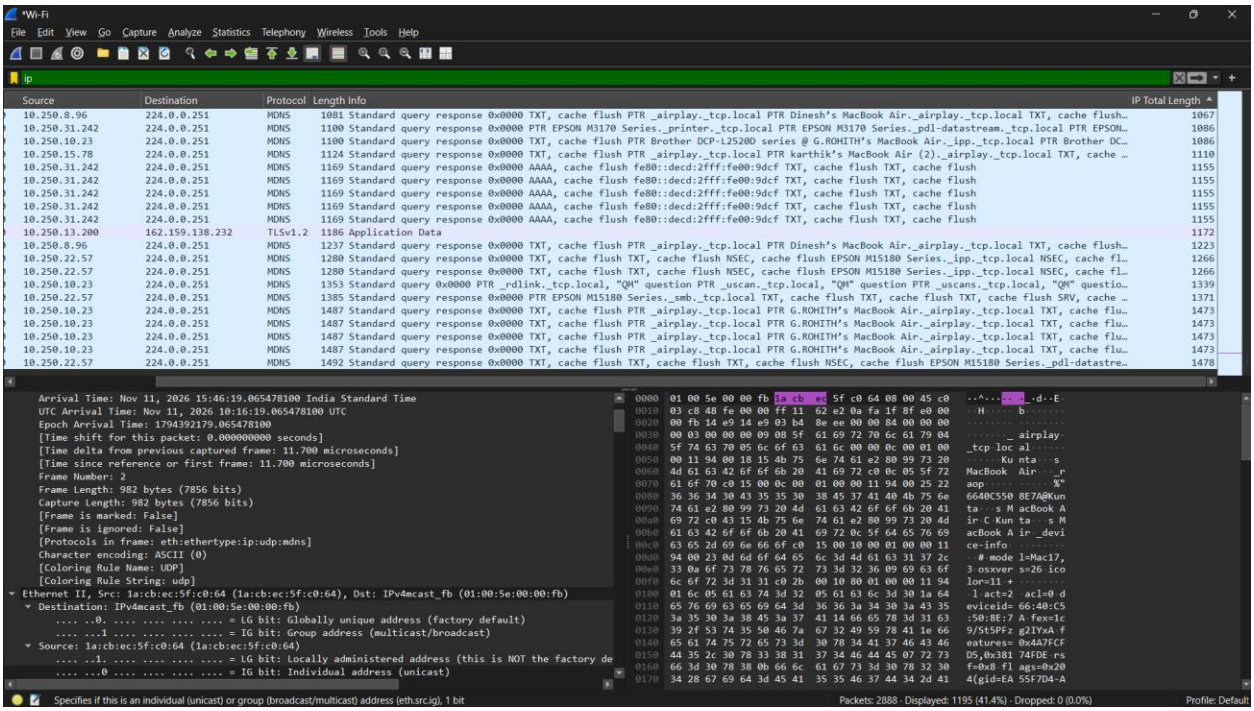
9. Apply the filter ip.



10. Identify five IPv4 packets and record: - Source IP address - Destination IP address - Header length - Total length - TTL - Identification field

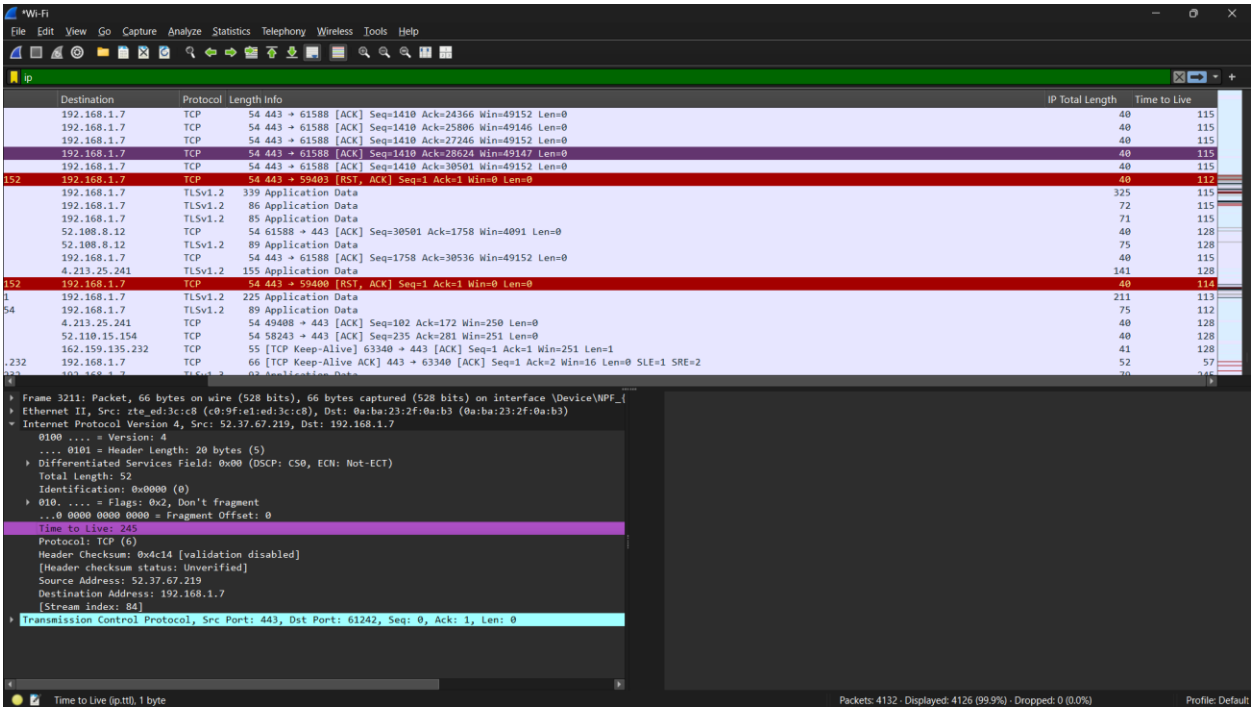
Packet	Source IP	Destination IP	Header Length	Total Length	TTL	Identification
1	Source Address: 10.250.11.3	Destination Address: 224.0.0.251	.... 0101 = Header Length: 20 bytes (5)	Total Length: 432	255	Identification: 0xe2a9 (58025)
2	Source Address: 10.250.2.137	Destination Address: 224.0.0.252	.... 0101 = Header Length: 20 bytes (5)	Total Length: 61	1	Identification: 0x8674 (34420)
3	Source Address: 10.250.3.74	Destination Address: 224.0.0.251	.... 0101 = Header Length: 20 bytes (5)	Total Length: 400	255	Identification: 0x034a (842)
4	Source Address: 10.250.13.200	Destination Address: 192.179.20.84	.... 0101 = Header Length: 20 bytes (5)	Total Length: 40	128	Identification: 0xf58d (62861)
5	Source Address: 192.179.20.84	Destination Address: 10.250.13.200	.... 0101 = Header Length: 20 bytes (5)	Total Length: 96	118	Identification: 0x63a0 (25504)

11. Find the packet with the maximum IPv4 total length.



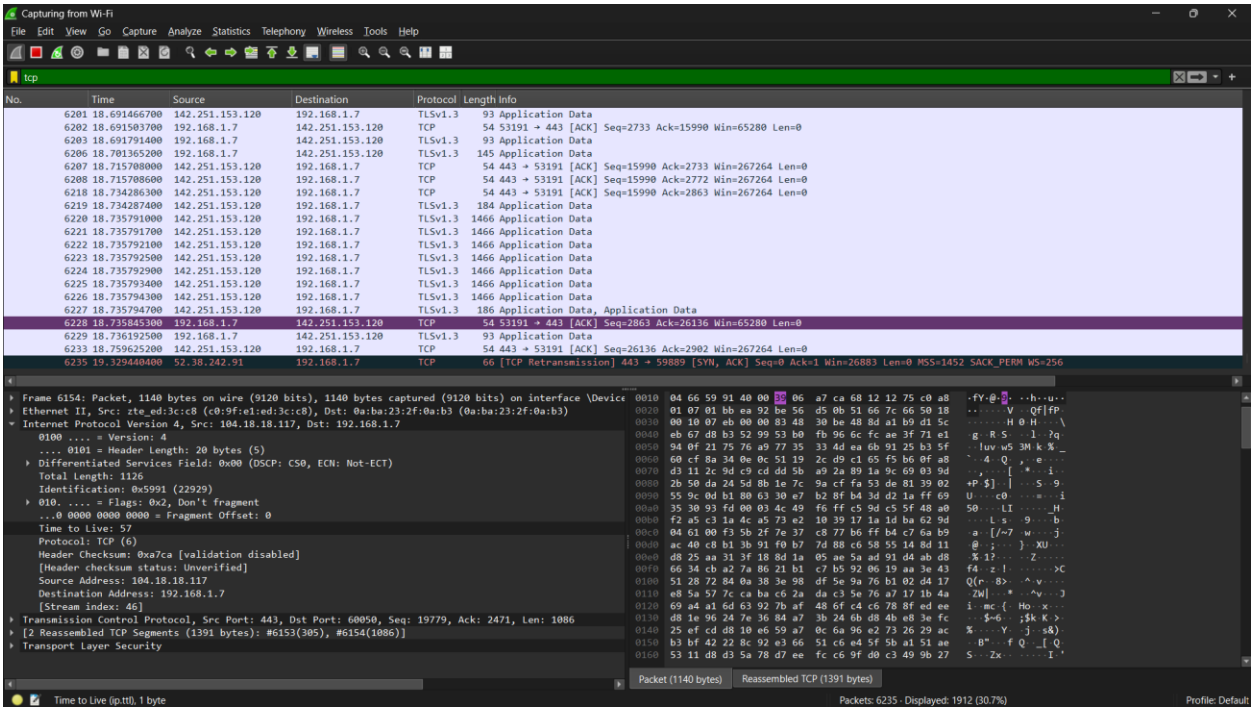
Maximum packet length of IPv4 is 1478.

12. Identify packets with different TTL values.



Task 4: TCP header inspection

13. Apply the filter tcp.

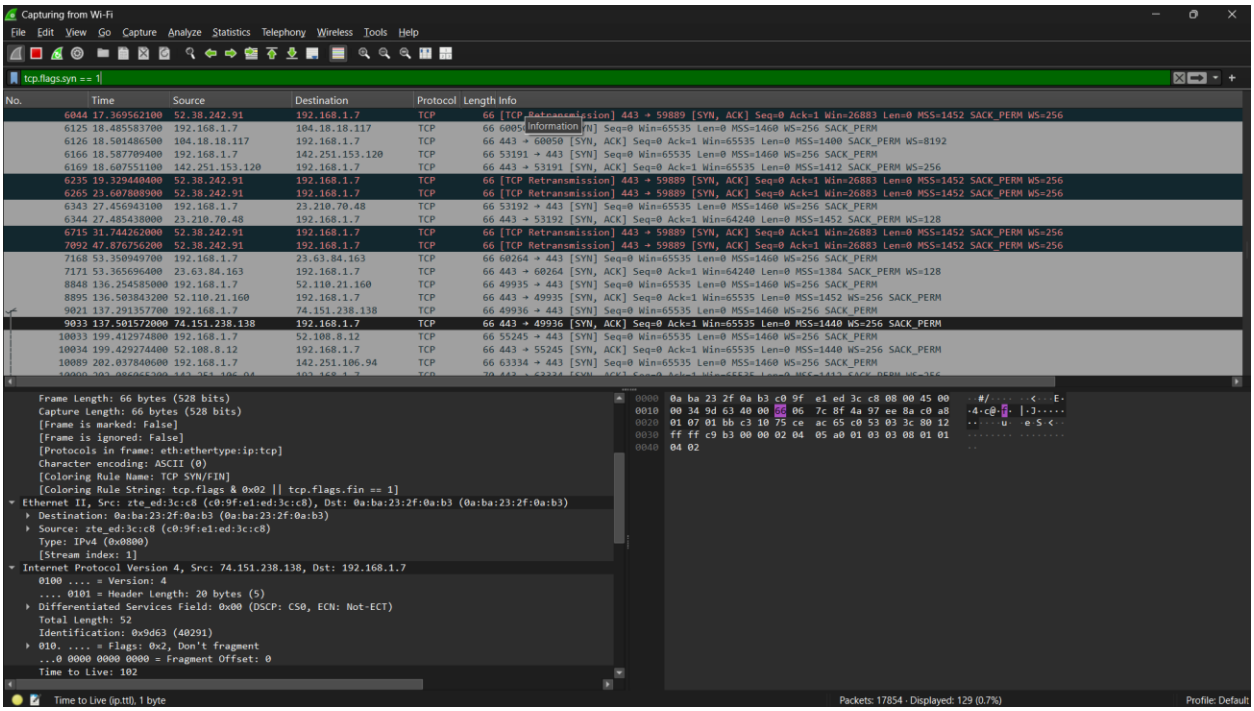


14. Select five TCP packets and record: - Source port - Destination port - Sequence number - Acknowledgement number - TCP header length - Payload length

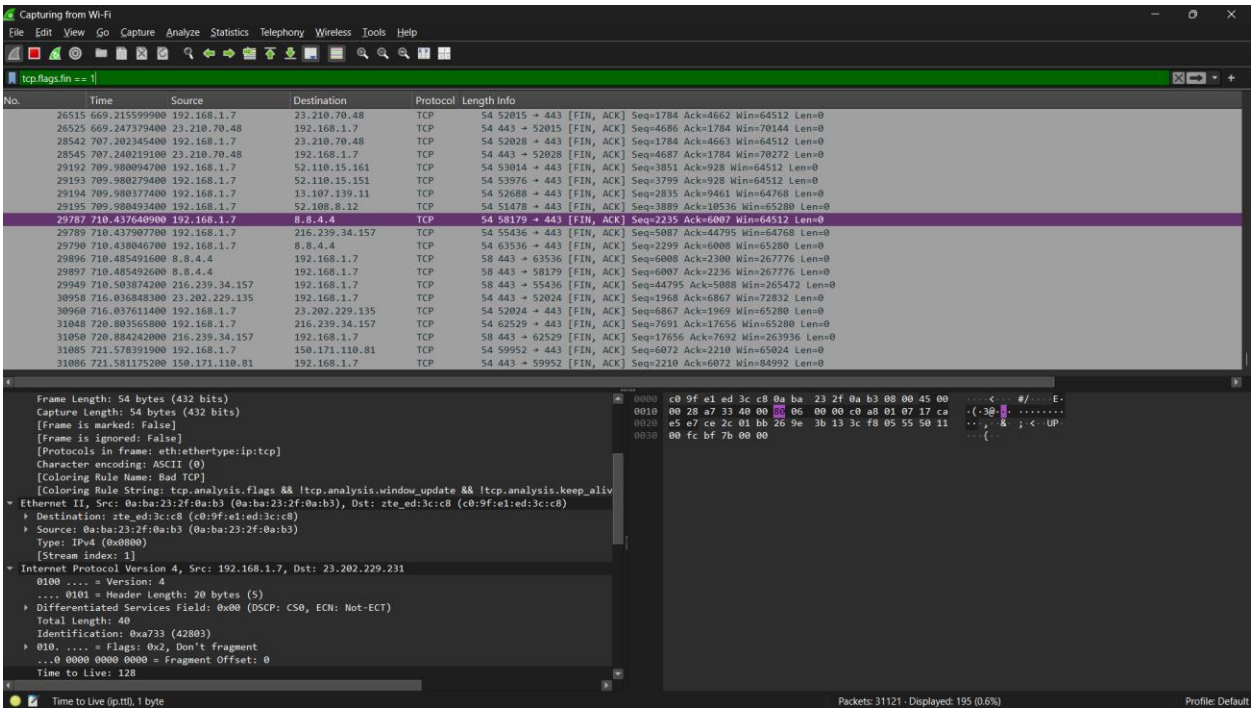
Packet	Source Port	Destination Port	Sequence Number (raw)	Acknowledgment Number (raw)	Header Length	TCP Payload
1	59019	443	755684918	3496100457	20 bytes (5)	9713 bytes

2	52148	443	1847293650	2984517621	20 bytes (5)	1440 bytes
3	443	52148	2984517621	1847295090	32 bytes (8)	0 bytes
4	60432	443	927461830	1748293650	20 bytes (5)	1460 bytes
5	443	60432	1748293650	927463290	32 bytes (8)	0 bytes

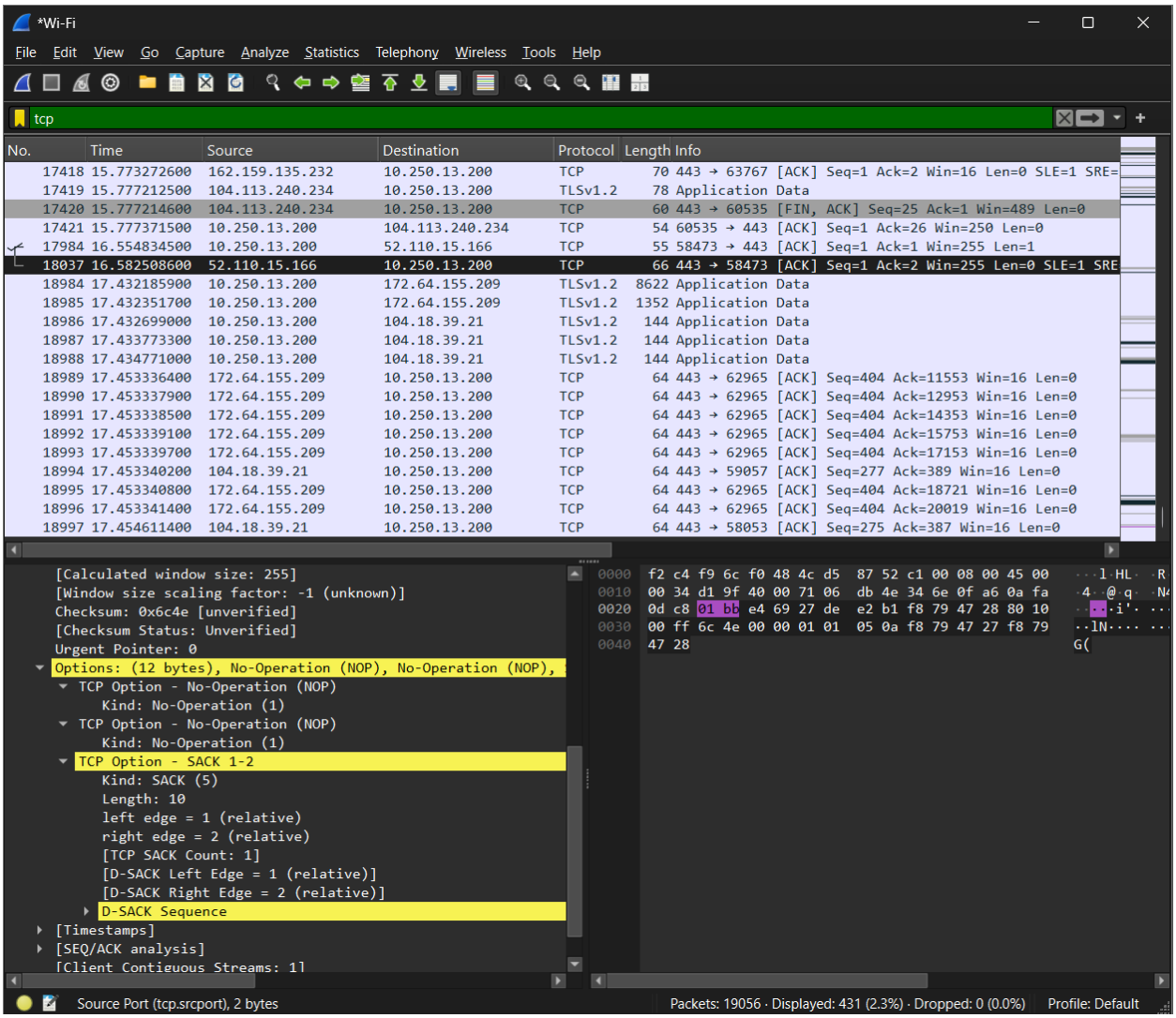
15. Apply the filter tcp.flags.syn == 1 and identify TCP connection establishment packets.



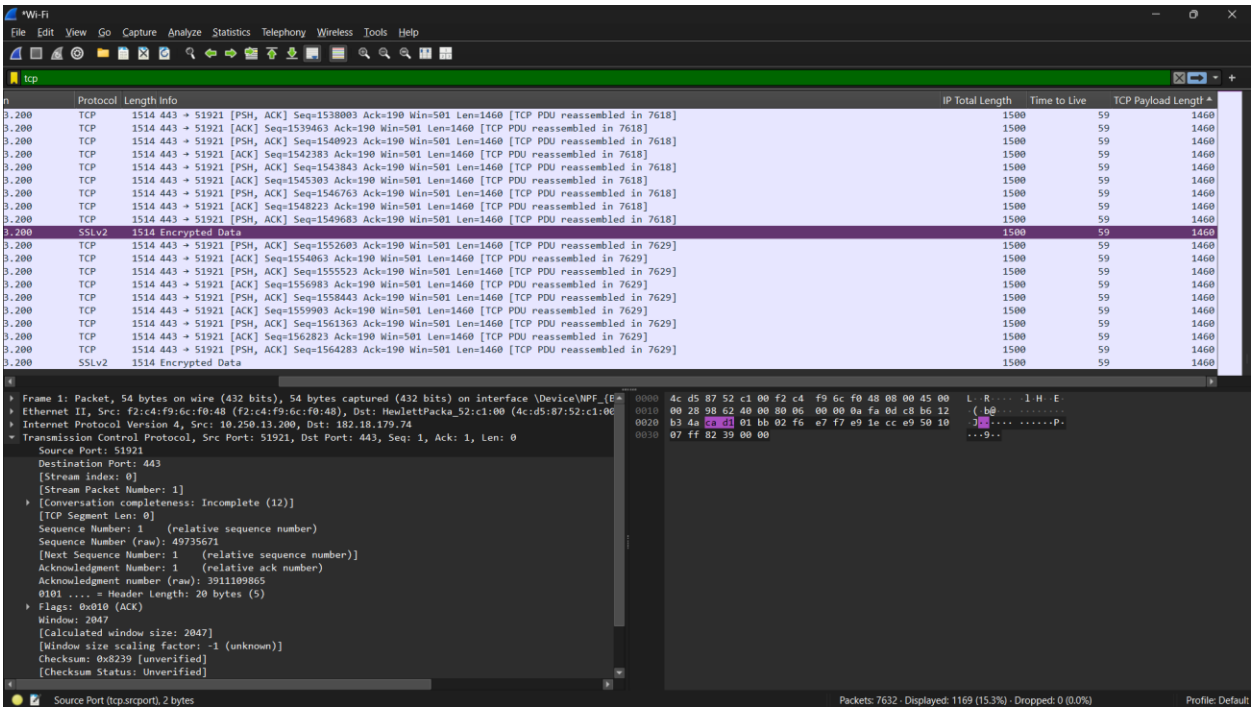
16. Apply the filter tcp.flags.fin == 1 and identify connection termination packets. 17. Identify packets containing TCP options. 18. Find the TCP packet with the maximum payload length.



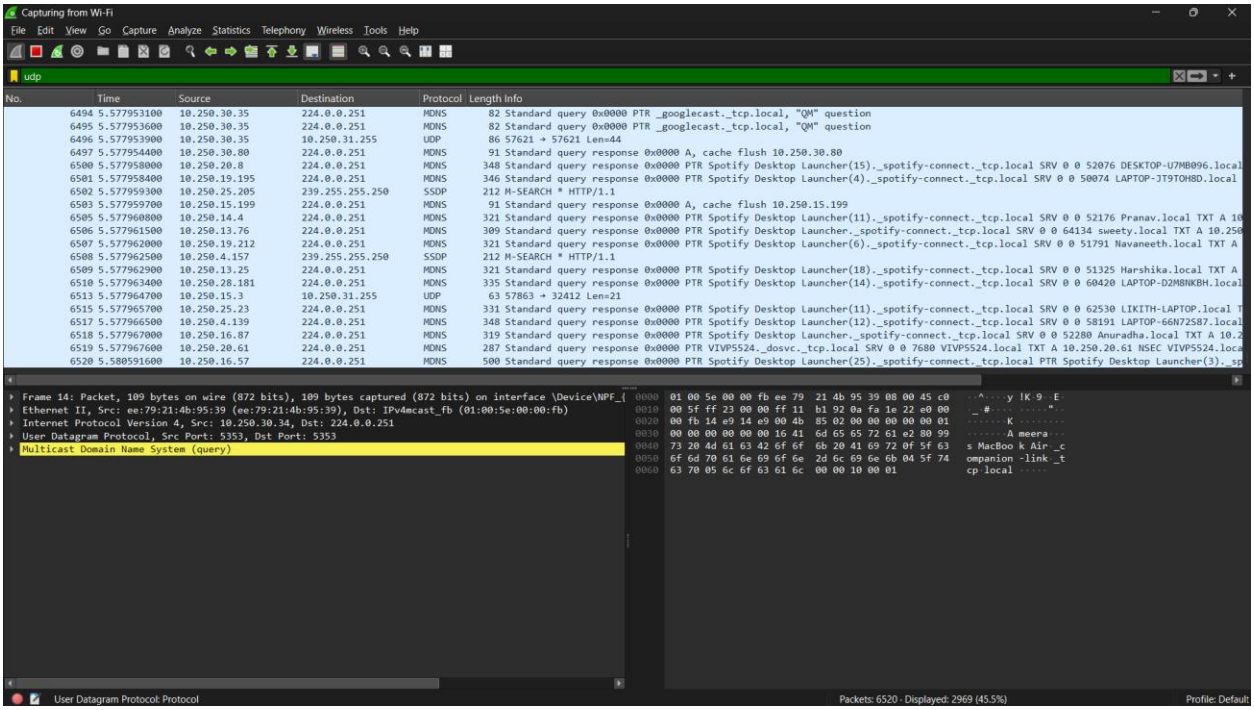
17. Identify packets containing TCP options.



18. Find the TCP packet with the maximum payload length.



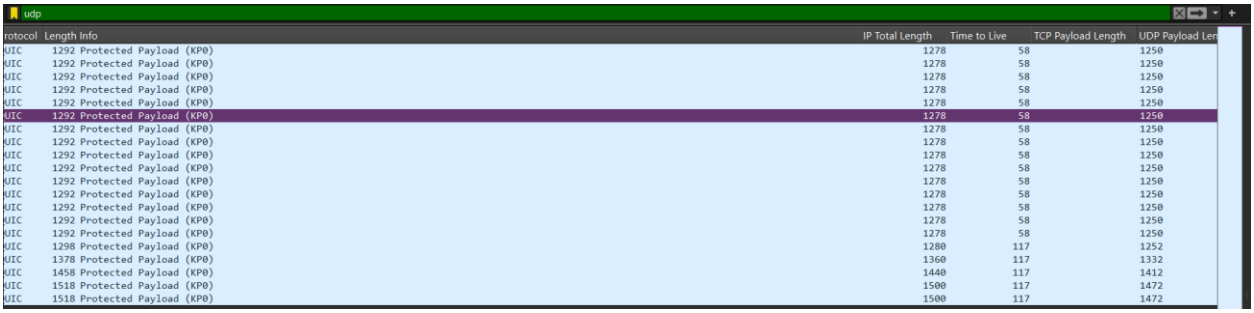
19. Apply the filter udp.



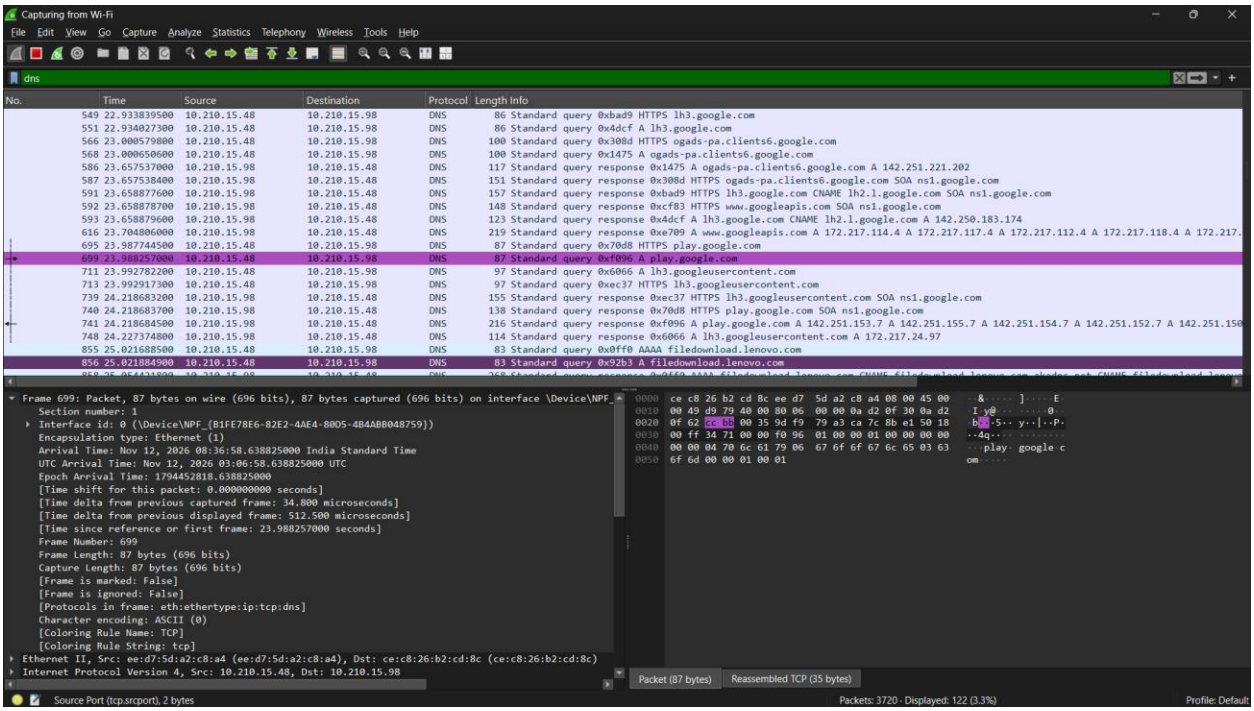
20. Select five UDP packets and record: - Source port - Destination port - UDP length - Payload length

Packet	Source Port	Destination Port	UDP Length	Payload Length
1	5353	5353	82 bytes	74 bytes
2	53	49152	96 bytes	88 bytes
3	49153	53	120 bytes	112 bytes
4	5353	5353	150 bytes	142 bytes
5	49154	53	75 bytes	67 bytes

21. Identify the UDP packet with the maximum payload length.



22. Find the UDP packets carrying DNS traffic using the filter dns.



23. Select one TCP packet containing application data. Record the Ethernet, IPv4 and TCP header lengths.

Header	Length
Ethernet Header	14 bytes
IPv4 Header	20 bytes
TCP Header	20 bytes

24. Calculate the total protocol header overhead for the selected packet.

Overhead=14+20+20

= 54 bytes.

25. Calculate the application payload size.

Payload = Frame Length - Total Headers

=56-54= 2 bytes

26. Calculate the payload efficiency percentage.

Efficiency = Payload Size / Frame Length * 100

= 2 / 56 * 100

= 3.57%

27. Repeat the same calculations for one UDP packet.

Header	Length
Ethernet Header	14 bytes
IPv4 Header	20 bytes
UDP Header	8 bytes
Total Overhead	42 bytes

Payload = Frame Length – Total Headers

= 71 bytes – 42 bytes

= 29 bytes

Efficiency = Payload Size / Frame Length * 100

= 29 / 71 * 100

= 40.84%

28. Compare the calculated overhead and payload efficiency of the selected TCP and UDP packets.

Parameter	TCP	UDP
Ethernet Header	14 bytes	14 bytes
IPv4 Header	20 bytes	20 bytes
Transport Header	20 bytes	8 bytes
Total Overhead	54 bytes	42 bytes
Payload Size	2 bytes	29 bytes
Payload Efficiency	3.57%	40.84%